

DIGITAL FM RECEIVER DESIGN USING MICROCONTROLLER AND I²C INTERFACING TECHNIQUES

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sections of prototype and their functioning are explained in this paper.

Abstract: The FM radio reception can be improved by Digital Processing and Intelligent Tuning Systems. The quality of FM radio reception can be further improved with the help of digital tuning and digital signal processing techniques. This research aims to use a combination of embedded system and digital radio receiver module to design a prototype for better FM reception and tuning which is easy to implement, cost effective and adaptable. This project is based on Philips TEA5767 digital radio receiver module. TEA5767 module has features such as stereo or mono outputs, radio station scanning and signal level indication. The Arduino Uno Development board is used to interface the TEA5767 Module and to send the commands for receiving FM signal. The main advantage of using TEA5767 IC is that it supports digital tuning, by giving read command, values from the TEA5767 module are read by the microcontroller. Similarly write command is provided by the microcontroller for scanning frequency range, after which audio playback can be started. Digital tuning methods enhances overall system functioning by providing better selectivity, sensitivity and accurate tuning. The proposed radio receiver is stable in terms of frequency tuning and requires a minimum of small and low cost external components.

1. Introduction

With the advancement in digital technology the use of microcontroller for communication became very popular. As it is observed that FM receiver of analog tuning method is difficult to tune and various environmental effects changes the tuned frequency of FM receiver. In order to overcome the frequency variation problem, various digital tuning methods have evolved. The use of microcontroller makes it very easy to overcome the above problem by interfacing the FM receiving module with microcontroller to give commands to the TEA5767 module. Better tuning and reception is achieved by using phase locked loop (PLL), TEA5767 is connected with microcontroller for giving control commands to module. The module will act according to the commands given by controller. The method of giving commands is through I²C. The module will give acknowledgements and other useful data to the microcontroller. The audio output of the module is given to the LM386 based amplifier section and to a loudspeaker. Various

1.1 Philips TEA5767 FM Receiver Module

The TEA5767 is a simple stereo FM receiver chip. This new generation low voltage FM radio provides a fully integrated intermediate frequency (IF) selectivity and demodulation. This TEA5767 receiver module does not require any coalition (alignment), [2]. The basis of digital tuning is conventional PLL (phase locked loop) concept [3]. The characteristics of TEA5767 IC are given below [8].

- High sensitivity due to integrated low-noise Radio Frequency input amplifier.
- RF Automatic Gain Control circuit.
- Phase-locked loop (PLL) synthesizer tuning system.
- I²C-bus and 3-wire bus, selectable via pin BUSMODE.
- 7-bit IF counter output via the bus.
- 4-bit level information output via the bus.
- Adjustment-free stereo decoder.
- Autonomous search tuning function.
- Standby mode.
- Two software programmable ports.
- Bus-enable line to switch the bus input and output lines into 3-state mode.

Figure 1 shows the pin diagram of TEA5767 FM Receiver Module with all input and output connections labelling.

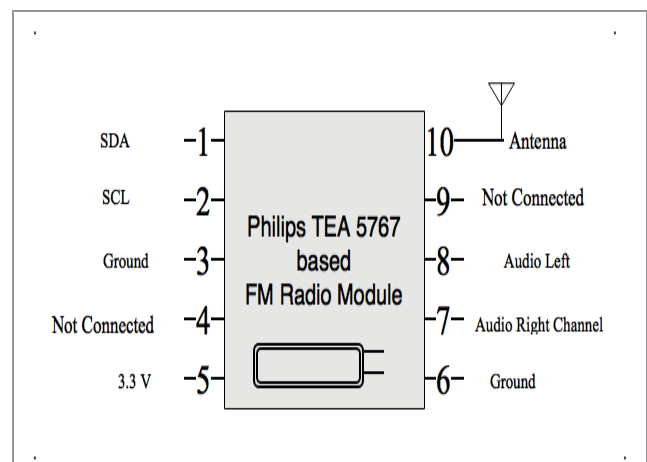


Figure 1 - TEA5767 Module Pin Diagram

1.2 Audio Amplifier Section

The LM386 consists of a low voltage power amplifier. It is basically used for battery-powered devices such as radios,

guitar amplifiers, and hobby electronics projects. In prototype LM386 based amplifier is used as the audio output from TEA5767 module is not enough to drive speakers. A preset is used for volume control.

1.3 Phase Locked Loop (PLL)

Phase-locked loops are widely used in the radio, telecommunications, computers and other electronic applications for locking the carrier frequency [3]. Basic PLL System is shown in Figure 2. It can be used in various devices to demodulate a signal, recover a signal from a noisy communication channel, generate an accurate and stable frequency at multiples of an input frequency, or distribute precisely timed clock pulses in digital logic circuits such as microprocessors related devices [4]. Since a single integrated circuit can provide a complete phase-locked-loop system, the technique is widely used in modern electronic devices.

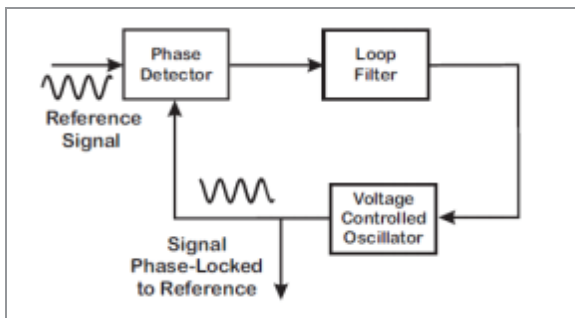


Figure 2 - Basic PLL System

1.4 Inter-Integrated Circuit (I²C)

I²C (Inter-Integrated Circuit), pronounced I-squared-C, is a multi-master, multi-slave, single-ended, serial computer bus developed by Philips Semiconductor. It is typically used for attaching lower-speed peripheral ICs to processors and microcontrollers for short-distance, intra-board communication [6]. I²C uses only two bidirectional open-drain lines, Serial Data Line (SDA) and Serial Clock Line (SCL), pulled up with resistors [5]. Typical voltages used are +5 V or +3.3 V although systems with other voltages are permitted. The connection of devices using I²C is shown in Figure 3. In the same figure there is one master device and three Slave Devices. In the project ATMEGA 328 / 8 microcontroller is used as master device and TEA5767 module(s) as slave device.

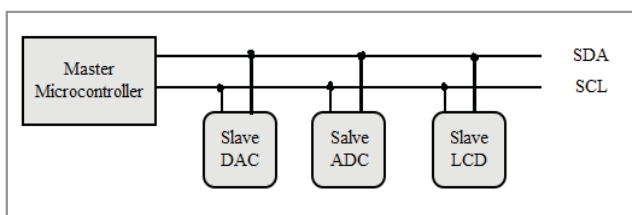


Figure 3 - Inter Integrated Circuit Connection

2. Implementation

The prototype is designed using ATmega328P Microcontroller based Arduino UNO development Board. The prototype is designed to function in auto scan mode in which a channel is searched automatically. The signal level is constantly monitored and the microcontroller is programmed to set receiver frequency accurately. The output from the TEA5767 Module is feed to amplifier section [1]. The antenna can be designed as per the application required. The communication between the microcontroller and TEA5767 module is using I²C methods. The main disadvantage of manual tuning is that it is

not accurate and various natural factors can change the frequency, proposed FM receiver overcomes problem faced in manual tuning receiver through digital tuning.

Figure 5 illustrates connection of FM Receiver module with the Arduino UNO board. Antenna is connected to pin 10 of the module and it catches the radio signals. The I²C pins of the module are SDA (pin 1) and SCL (pin 2), SDA is data pin and SCL is used for clock signal. The synchronisation and data transfer between module and microcontroller is performed using pin 1 and pin 2 of the module. In order to give audio output after demodulation of signal pin 8 and pin 7 of module is used. The pin 8 is audio output pin for left channel and pin 7 for right channel, pin diagram of TEA 5767 module is illustrated in figure 1. Signal from audio output pin(s) is provided as input to audio amplifier section in order to boost the signal to drive speakers.

Arduino UNO development board is used as platform to design prototype of the radio receiver, controlling of the module performed by using different development board with ATMEGA 8 / ATMEGA 16 / ATMEGA 328 microcontroller support. LCD module is connected to the digital pins present on the development board, values obtained from TEA5767 module are stored in microcontroller and displayed to the user through LCD. In this paper FM receiver using ATMEGA 8 microcontroller is also shown in Figure 8, which is implemented on customized PCB with less options and limited radio receiver functions.

3. Software

Program for tuning and scanning FM channel(s) is developed by free software integrated development environment (IDE) available at Arduino Website. The open-source Arduino Software IDE makes it easy to write code and upload it to the board. It runs on Windows, Mac OS, and Linux. This software can be used with any Arduino board [9]. After the software is installed successfully on personal computer, it can be used to develop applications specific program by connecting suitable sensors. In this project programming for I²C interface and LCD interfacing is carried out for frequency tuning and display controlling.

4. Process Flow

The algorithm for this project is as given

- Initialize the TEA5767 module.
- Write data to TEA5767 module.
- Read data from TEA5767 module.
- Display details on LCD.
- Scan & Monitoring of the available FM channels

The process flow chart in Figure 4 gives details about signal reception by prototype through the Program Code written using the Arduino IDE. It is illustrated in process flow chart that TEA 5767 module provides signal level information and can be used for monitoring availability of a signal and if broadcast is switched off then error message is displayed and the system will ask the user to switch different frequency. The receiver module information is read by the microcontroller during read cycle and stored in microcontroller's memory, after particular time intervals the read cycle is reinitiated to update records.

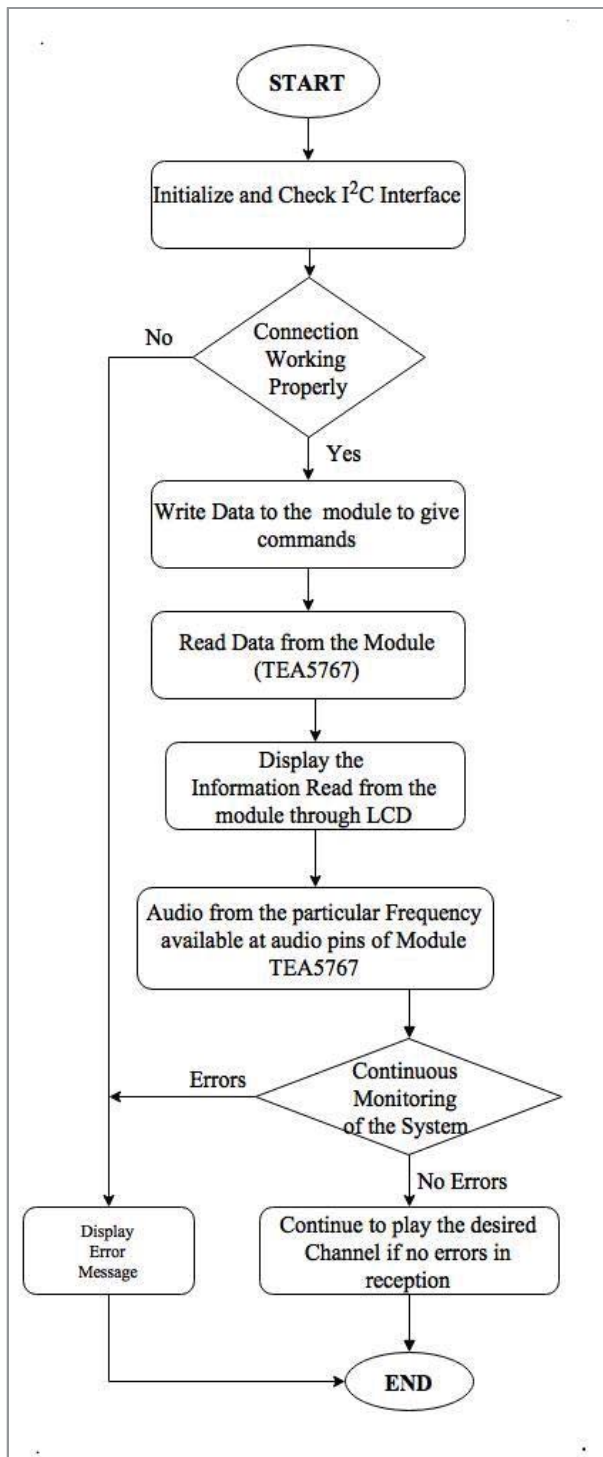


Figure 4 – Process Flow Chart

5. Working of FM Receiver Module

First initialization of TEA5767 module is done, after initialization of module commands for frequency searching are given to the module by microcontroller. Then the TEA5767 module provides acknowledgments and details about whether a station is found or not. The values of signal strength are also provided by the module [6]. We can also check the type of reception whether it is “stereo or mono”. The data obtained from TEA5767 module is displayed using LCD screen. The block diagram given below in Figure 5 illustrates the connection of the components in the project. In the project Amplifier section is connected to the audio output port of the TEA5767 IC. The Audio output is not processed through the Arduino Uno Board.

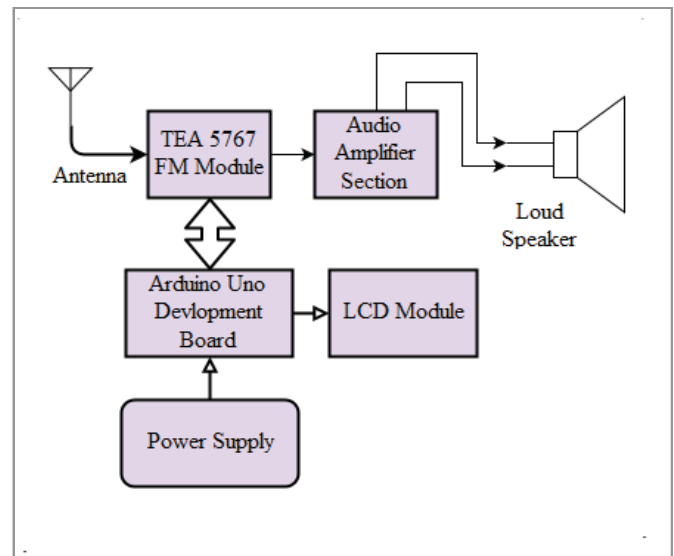


Figure 5 – Connection Diagram

6. Result

FM radio receiver is implemented using TEA5767 Module. The TEA5767 Module is successfully interfaced with Arduino UNO R3 development board. Using TEA5767 Module accurate frequency tuning can be done. The soldering work of wires with TEA5767 Module requires special tools. The signal received by the Module is resistant to various temperature fluctuations unlike the analog FM Receivers. Level in the LCD gives a indication about the signal level. We were able to implement auto scan feature in the system. The signal strength is compared by the chip to work in auto scan mode. Variations in reception are observed when the length of Antenna is changed.

7. Conclusion

This paper elaborates the usage of digital systems for FM radio reception using a microcontroller. The I²C communication method is used for sending the control commands to the TEA5767 FM Radio Module. The command data byte(s) are also sent to the microcontroller by the module to give details about the status of the FM reception like the STEREO or MONO reception, Frequency details and Auto Scan Status. The tuning of the receiver for a particular frequency can also be easily handled using the commands by the microcontroller.

The cost of a digital tuning FM receiver is reduced by using combination of microcontroller and FM receiver Module. Further the size of complete circuit can be even reduced by using an advance microcontroller with less number of pins having I²C options [7]. The design used in this project is useful in making FM Receiver using microcontroller and various FM Module. In future additional function(s) related to Radio Data System (RDS) can be added to the project. The study of FM Reception by changing the types of Antenna can be done.

The complete project can be made by using ATmega328P Microcontroller without using Arduino Uno Development Board in this case some additional components also required. A photograph of working FM Receiver is shown in Figure 6. The soldered TEA 5767 IC Module is shown in Figure 7, [10]. After designing receiver on development board it is also implemented using ATMEGA 8 microcontroller on a customised PCB illustrated in Figure 8.

Use of Microcontroller with FM Receiver Module provides excellent selectivity and sensitivity and at the same time better control on searching and tuning is achieved.

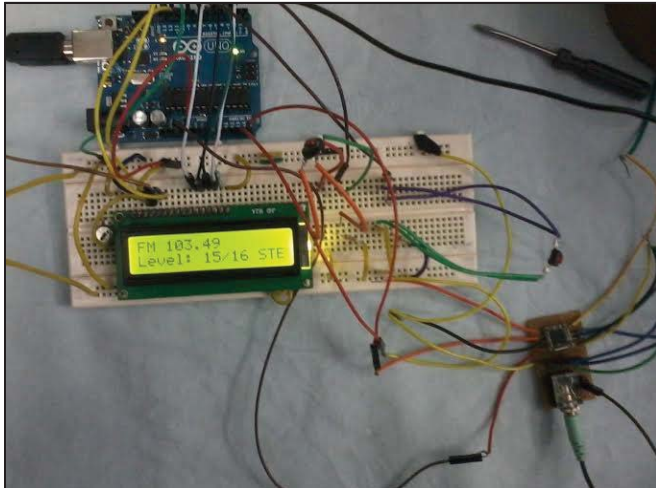


Figure 6 - TEA5767 FM Radio assembled on breadboard

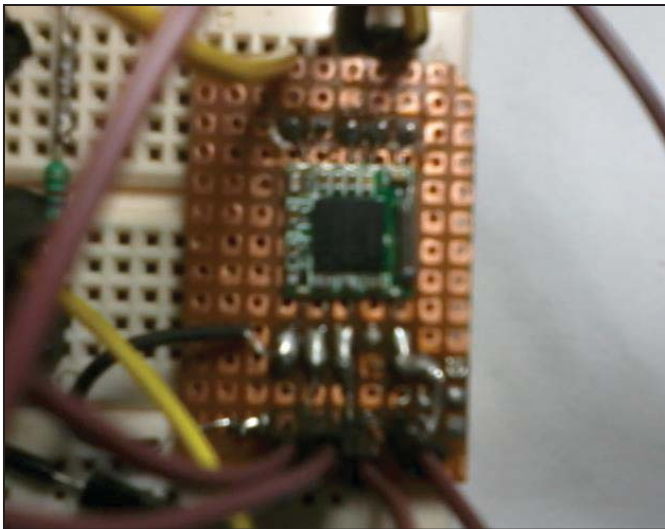


Figure 7 - TEA5767 Module Soldered on PCB

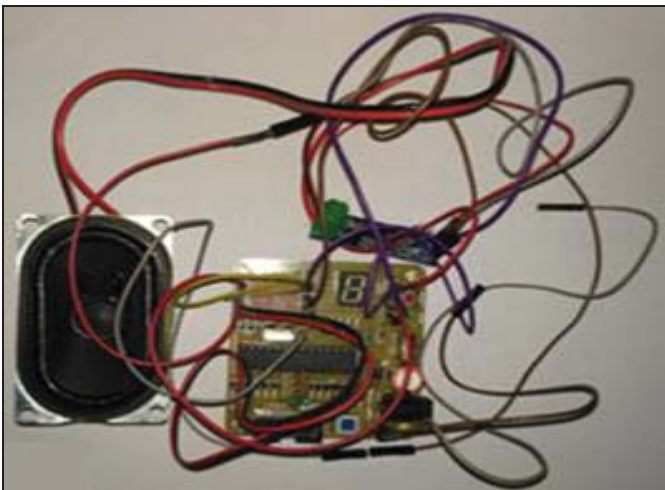


Figure 8 – Prototype on custom PCB using ATMEGA 8

8. References

[1.] Joseph P. Hoffbeck, Mark M. Sugiyama, “Real-time FM Radio for Teaching DSP and Communication Systems” School of Engineering University of Portland Portland, Oregon, USA, 2013 IEEE.

[2.] B. Razavi, “RF Microelectronics”. Englewood Cliffs, NJ: Prentice-Hall, 1998.

[3.] Mao Lai and Michino Nakano, “Special Section on Phase Locked Loop Techniques”, Guest Editorial, IEEE Transactions on Industrial Electronics, Vol. 43, No 6, December 1996.

[4.] F.M Gardner, “Phase Locked Loop Techniques”, 2nd edition; New York; Wiley 1979.

[5.] M. Schwartz, W. R. Bennett, S. Stein, “Communication systems and Techniques”. McGraw-Hill, 1966 (original publication); reprinted by IEEE Press, 1996.

[6.] Pankaj Kumar Mehto, Pragya Mishra, Sonu Lal, “Design and Implementation for Interfacing Two Integrated Device Using I2C Bus” IJRICCE, ISSN (Online): 2320- 9801, Vol. 2, Issue 3, March 2013.

[7.] K. Taura, et al., “A new approach to VHF/FM broadcast receiver using digital signal processing techniques,” IEEE Transactions on Consumer Electronics, Vol. 46, No. 3, pp. 751-757, June 2000.

[8.] NXP. 2007., “Low-power FM stereo radio for handheld applications,” TEA5767HN datasheet, Rev. 5 [Revised Jan 2007].

[9.] Atmel 2015 “Atmel 8-Bit Microcontroller with 4/8/16/32kbytes In-System Programmable Flash” ATMEGA 48A/ PA/ 88A/ PA/ 168A/ PA/ 328/ P Datasheet [Nov 2015].

[10.] Madhuran Mishra, “Arduino Based FM Receiver”, EFY July 2014, Page 88.